AQ1.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
5.1 Level 1:

1 point
Choose the set of correct options

- [ ] If A is a real 3×3 matrix, then $det(A) = det(A^T)$.
- [ ] If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 4×4.
- [ ] If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 3×3.
- [ ] If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are $1 \times 1, 2 \times 2,$ and 3×3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If A is a real 3×3 matrix, then $det(A) = det(A^T)$.
If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 3×3.
If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are $1 \times 1, 2 \times 2,$ and 3×3.

1 point
Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a 3×3 matrix. Which of the following is(are) correct?

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- [ ] $det(A) = 3 \times det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) - 2 \times det\left(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$
- [ ] $det(A) = 3 \times det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$
- [ ] $det(A) = -2 \times det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 3 \times det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) - 2 \times det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right)$
- [ ] $det(A) = 2 \times det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) - 3 \times det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right)$

$+ 2 \times \det\left(\begin{bmatrix}3&2\\2&1\end{bmatrix}\right)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$det(A) = 3 \times det\left(\begin{bmatrix}3&2\\1&1\end{bmatrix}\right) - 2 \times det\left(\begin{bmatrix}2&2\\2&1\end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix}2&3\\2&1\end{bmatrix}\right)$

$det(A) = 3 \times det\left(\begin{bmatrix}3&2\\1&1\end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix}2&2\\1&2\end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix}2&3\\2&1\end{bmatrix}\right)$

$det(A) = -2 \times det\left(\begin{bmatrix}2&2\\1&1\end{bmatrix}\right) + 3 \times det\left(\begin{bmatrix}3&2\\2&1\end{bmatrix}\right) - 2 \times det\left(\begin{bmatrix}3&2\\2&1\end{bmatrix}\right)$

1 point

If all the elements of a 3×3 real matrix A are the same, then which of the following is (are) correct?

- [ ] Determinant of matrix A is 0.
- [ ] Determinant of matrix A cannot be determined from the given information.
- [ ] Determinant of matrix A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.
- [ ] Determinant of matrix $A + A^T$ cannot be determined from the given information, where A^T denotes the transpose of A.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Determinant of matrix A is 0.
Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.

1 point

Choose the set of correct options

- [ ] If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
- [ ] If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
- [ ] If one row is a non-zero multiple of another row in a 3×3 real matrix, then the determinant of that matrix is not zero.
- [ ] If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant of that matrix is zero.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant

of that matrix is zero.

Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ be a matrix of order 4×4. Find $det(A)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -1
1 point

5.2 Level 2:

1 point
Let $A, B,$ and C be 3×3 real matrices. Which of the following options is (are) correct?

- [ ] $det(ABC) = det(A)det(B)det(C)$
- [ ] $det(A^3) = det(A)^3$
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$
- [ ] $det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(ABC) = det(A)det(B)det(C)$
$det(A^3) = det(A)^3$
$det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.
1 point
Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i > j \\ j & \text{if } i \leq j \end{cases}$

Choose the set of correct options

[Hint: Write down matrix A first.]

- [ ] A is an upper triangular matrix.
- [ ] A is a lower triangular matrix.
- [ ] $det(A) = 6$
- [ ] $det(A) = 3$

No, the answer is incorrect.
Score: 0

Accepted Answers:
A is an upper triangular matrix.
$det(A) = 6$

1 point

Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i < j \\ 2 & \text{if } i \geq j \end{cases}$

Choose the set of correct options

- [ ] A is an upper triangular matrix.
- [ ] A is a lower triangular matrix.
- [ ] $det(A) = 2$
- [ ] $det(A) = 8$

No, the answer is incorrect.
Score: 0
Accepted Answers:
A is a lower triangular matrix.
$det(A) = 8$

Suppose $A = \begin{bmatrix} 4 & -1 & 3 \\ 2 & 0 & 1 \\ 3 & -2 & 0 \end{bmatrix}$ and C_{ij} is the (i, j)-th cofactor of the matrix A.

Let B be 3×3 matrix where (i, j)-th entry is equal to C_{ij}. Find $det(B)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 49

1 point

Suppose A and B are two 3×3 matrices such that $det(A) = 4$ and $B = 3A$. Find the value of $\sqrt[3]{det(A^2 B)}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 12

1 point

Check Answers

Your score is: 0/10